Quasi-static brittle fracture

Analysis of Francfort-Marigo model

Luca Pignatelli

Radboud Universiteit

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Historical remarks on LEFM

Griffith's criterion

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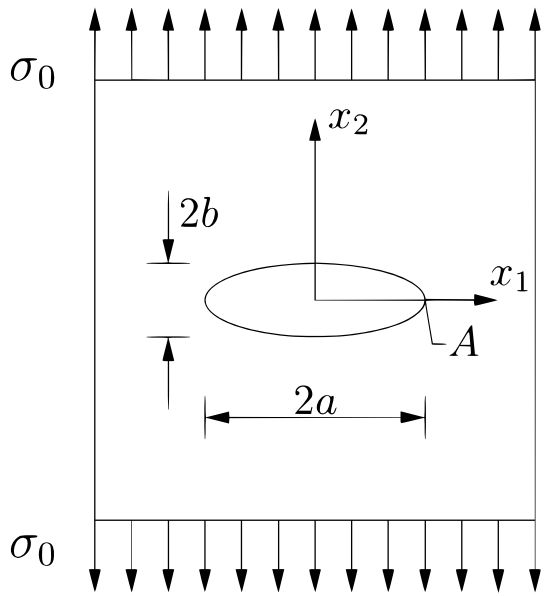

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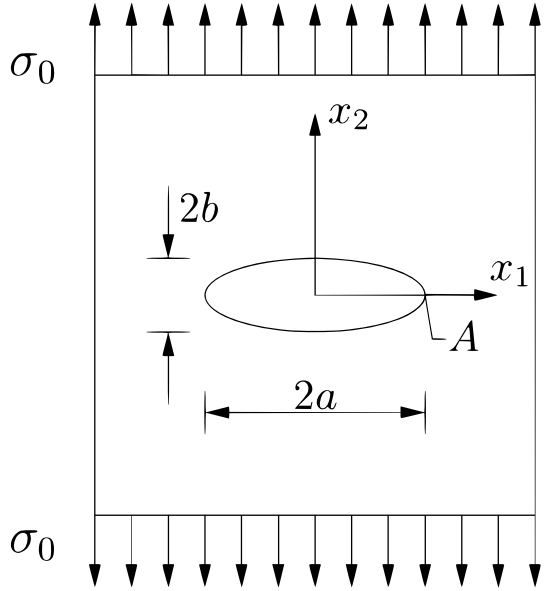

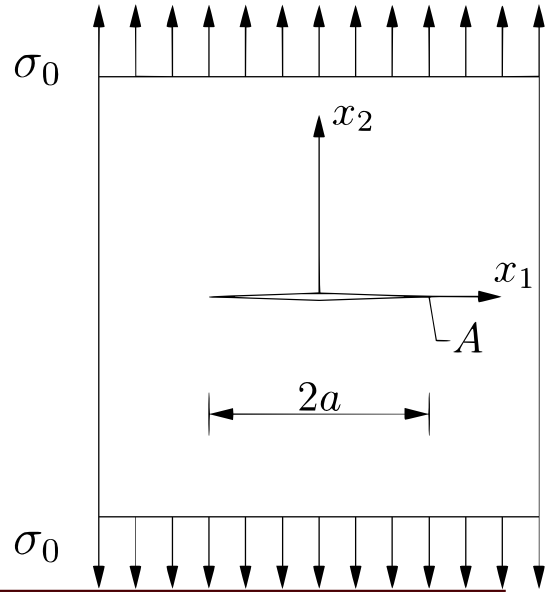

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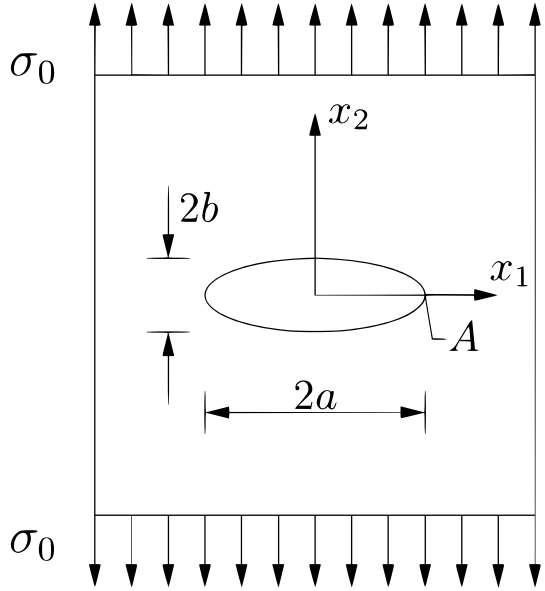
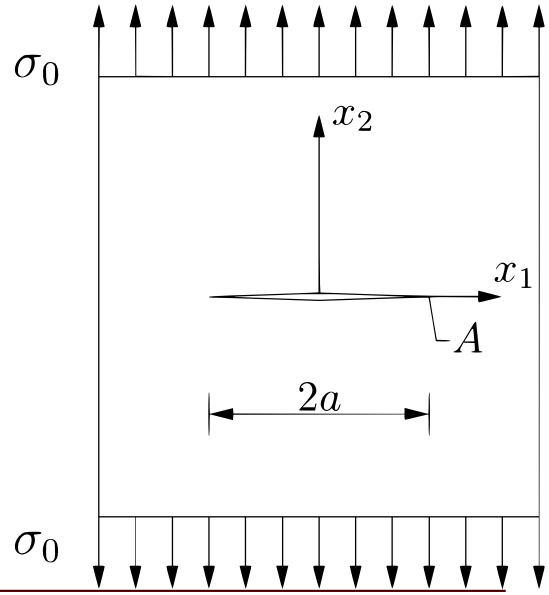


- Contribution of elastic, potential, and fracture energy

$$E_t(\ell) = E_e(\ell) + E_p(\ell) + E_f(\ell)$$

$$\underbrace{-\frac{1}{2}\left(\frac{\mathrm{d}E_e}{\mathrm{d}\ell} + \frac{\mathrm{d}E_p}{\mathrm{d}\ell}\right)}_{=:G} = \underbrace{\frac{1}{2}\frac{\mathrm{d}E_f}{\mathrm{d}\ell}}_{=:G_c}.$$



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- what happens then $G > G_c$?

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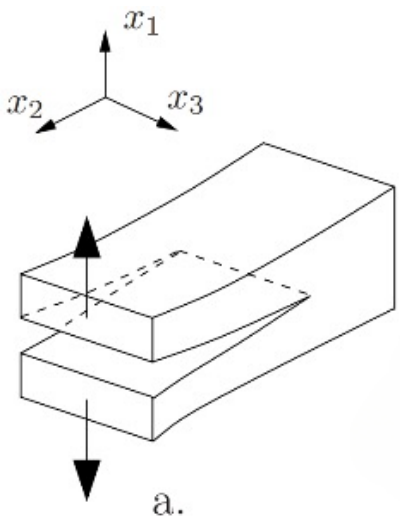

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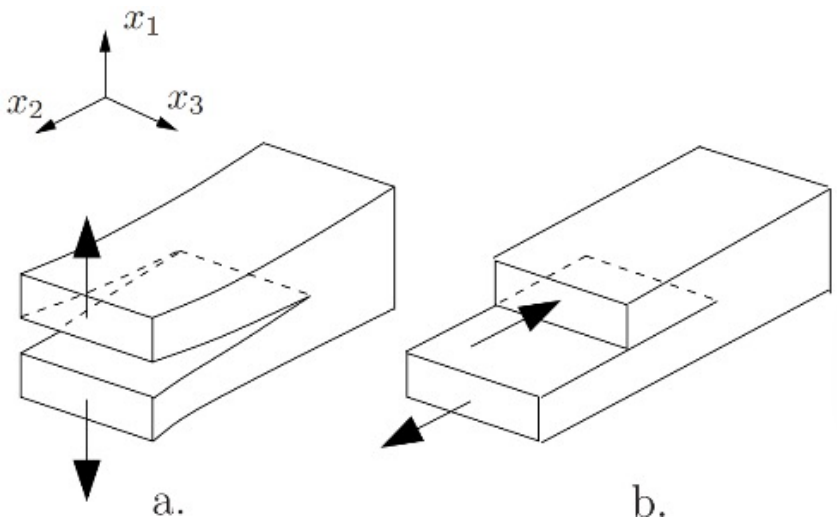

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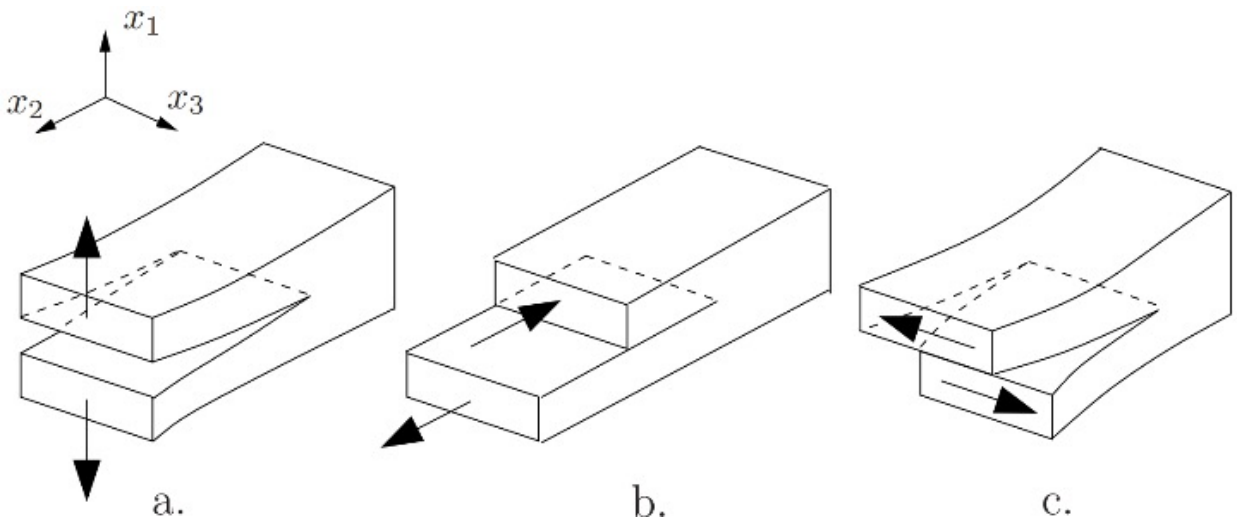

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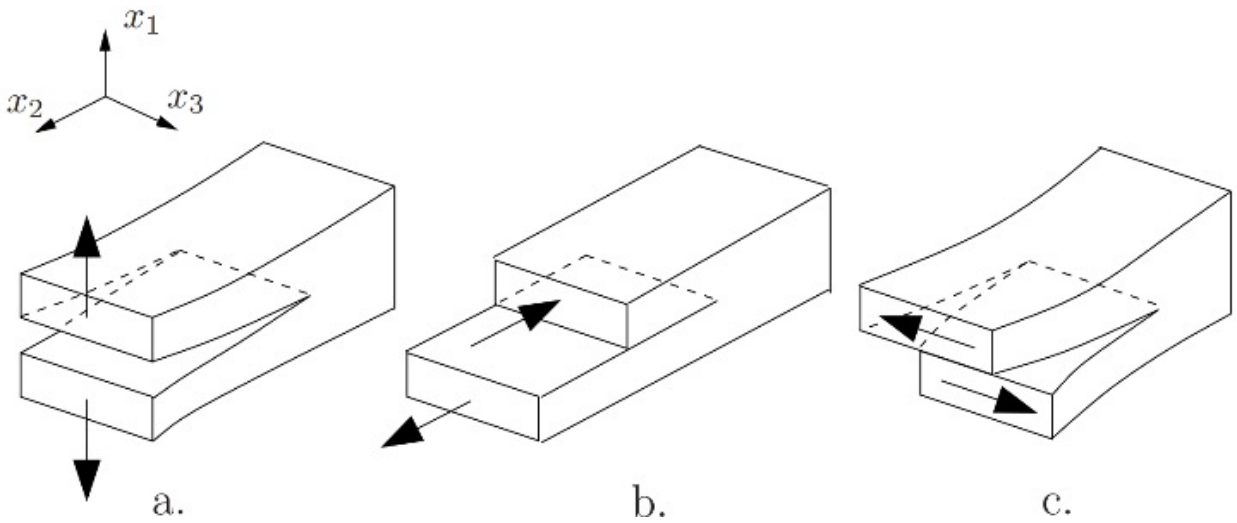

$$\sigma_{ij} = \frac{K_I}{\sqrt{r}} f_{ij}^{I}(\theta) + \frac{K_{II}}{\sqrt{r}} f_{ij}^{II}(\theta) + \frac{K_{III}}{\sqrt{r}} f_{ij}^{III}(\theta)$$

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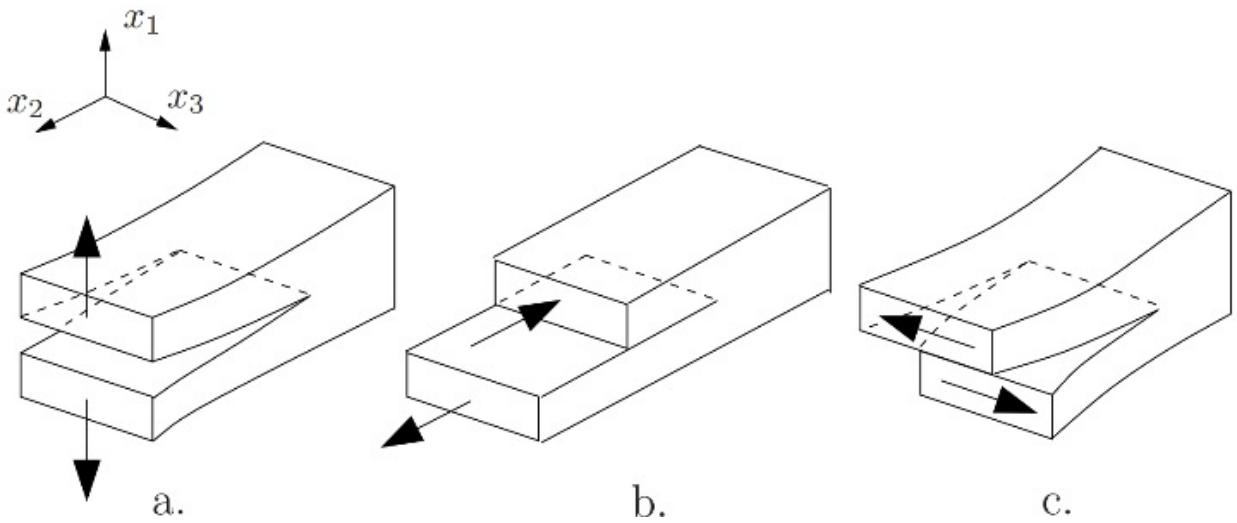

$$\sigma_{ij} = \frac{K_I}{\sqrt{r}} f^I_{ij}(\theta) + \frac{K_{II}}{\sqrt{r}} f^{II}_{ij}(\theta) + \frac{K_{III}}{\sqrt{r}} f^{III}_{ij}(\theta) + \sum_{n=0}^{\infty} A_n\, r^{\frac{n}{2}}\, g^n_{ij}(\theta)$$

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Usually $K_{Ic} < K_{IIc}, K_{IIIc} \implies$ mode I controls propagation.

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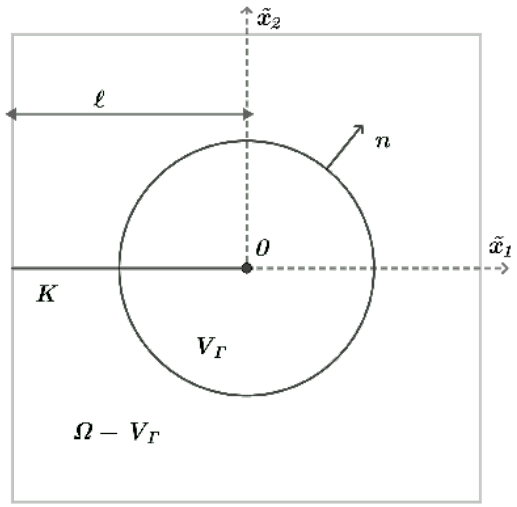

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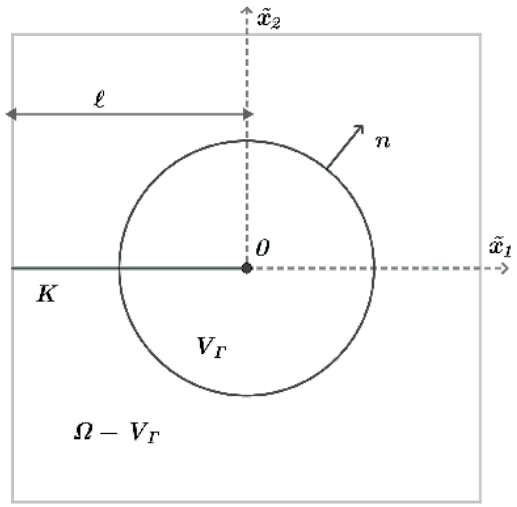

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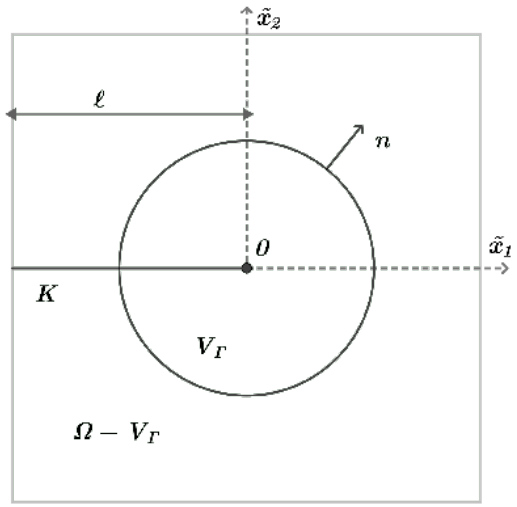

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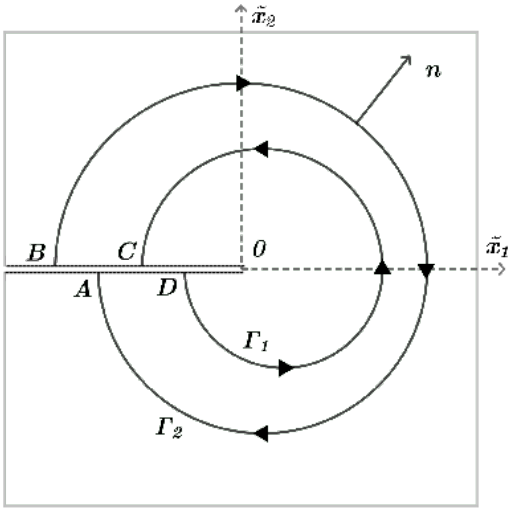

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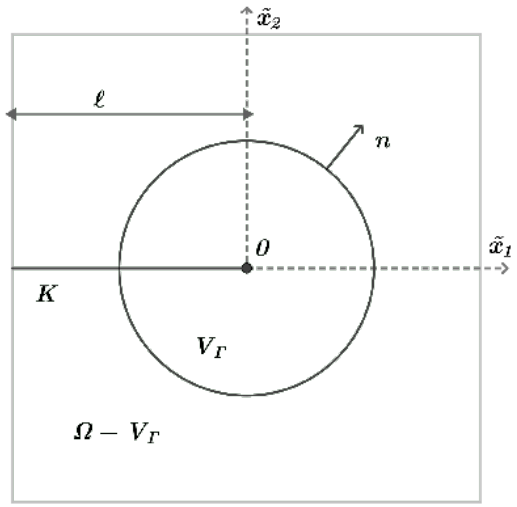

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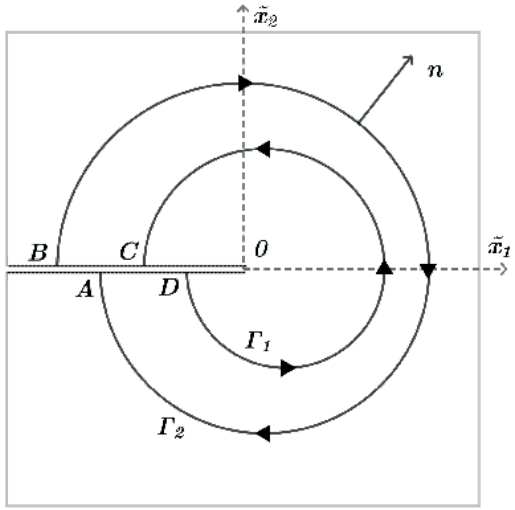
- $\mathrm{J} = G$, used to compute EER (in LE)



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Irreversible quasi-static Griffith's criteria

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Has the same limitations as Griffith's original criteria, but can be used with non-linear elasticity (with J).

Variational formulation

Francfort-Marigo model

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Objective: overcome limitations of classical LEFM.

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Variables:

- loading g;
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$$\mathcal{E}(v, K) := \underbrace{\int_{\Omega \setminus K} W(x, \varepsilon(v)(x))\, \mathrm{d}x}_{\text{Elastic energy}}$$

$\varepsilon(v) = \frac{1}{2}(\nabla v + \nabla^\top v)$.

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$$\mathcal{E}(v, K) := \int_{\Omega \setminus K} W(x, \varepsilon(v)(x))\,\mathrm{d}x + \underbrace{\int_K k(x)\,\mathrm{d}\mathcal{H}^{N-1}(x)}_{\text{Fracture energy}}$$

$\mathcal{H}^{N-1} = N - 1$-dimensional Hausdorff measure.

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Functions $u \in L^p_{\mathrm{loc}}$ with $\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla^\top u) \in L^p$.

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- Stationary energy for a given loading:

$$\frac{\mathrm{d}}{\mathrm{d}s} E(g(t), K(s))\Big|_{s=t} = 0 \qquad \text{for a.e. } t.$$

Strengths, weaknesses and extensions

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- **Weaknesses**
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Two-dimensional anti-plane linear elasticity

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- Constant fracture toughness $k(x) \equiv \gamma$:

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Interplay and minimization

$$E(g,K) := \min_{v \in \mathcal{V}(g,K)} \left\{ \frac{\mu}{2} \int_{\Omega \setminus K} |\nabla v|^2 \, \mathrm{d}x + \gamma \, \mathcal{H}^1(K) \right\};$$

$$\mathcal{V}(g,K) := \left\{ v \in L^{1,2}(\Omega \setminus K) : v = g \ \text{ q.e. on } \partial_D\Omega \setminus K \right\}.$$

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$$\frac{\mathrm{d}}{\mathrm{d}s} E(g(t), K(s))\Big|_{s=t} = 0 \quad \text{for a.e. } t \in [0, 1]$$

Preliminaries and overview of proof

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- If Ω is bounded with a Lipschitz boundary, then $L^{1,2}(\Omega) = H^1(\Omega)$.

Lower dimensional measures

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Let $\mathcal{A}_\varepsilon$ be a ε-fine covering of E. We define the s-dimensional Hausdorff measure of E as:

$$\mathcal{H}^s(E) := \sup_{\varepsilon \to 0} \mathcal{H}^s_\varepsilon(E) = \sup_{\varepsilon \to 0} \inf_{\mathcal{A}_\varepsilon} \sum_{A \in \mathcal{A}_\varepsilon} \alpha(s) \left(\operatorname{diam}(A)\right)^s.$$

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p-Capacity

Let $\Omega \subset \mathbb{R}^N$ be a bounded open set, and $A \subset \Omega$ an arbitrary subset. We define the p-Capacity of A in Ω as:

$$\operatorname{Cap}_p(A;\Omega) := \inf_{u \in \mathcal{W}(A;\Omega)} \int_\Omega |\nabla u|^p \, \mathrm{d}x,$$

$$\mathcal{W}(A;\Omega) := \left\{ u \in W^{1,p}_0(\Omega) \cap C(\Omega) : A \subseteq \{u \geq 1\}^\circ \right\},$$

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- if $1 < p \leq N$, then $\mathcal{H}^{N-1}(E) = 0$ implies $\mathrm{Cap}_p(E;\Omega) = 0$.

Hausdorff distance and lower semi-continuity

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The set $\mathcal{K}(\overline{\Omega})$ of compact non-empty subsets of $\overline{\Omega}$ is complete when equipped with the distance:

$$\mathcal{D}(K_1, K_2) := \max\left\{ \sup_{x \in K_1} d(x, K_2),\ \sup_{y \in K_2} d(y, K_1) \right\}.$$

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$\{K_n\}_{n \in \mathbb{N}}, K \subset \bar{\Omega}$ connected compact sets such that $\mathcal{D}(K_n, K) \to 0$. Then

$$\mathcal{H}^1(K) \leq \liminf_{n \to \infty} \mathcal{H}^1(K_n).$$

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Behavior near the tip

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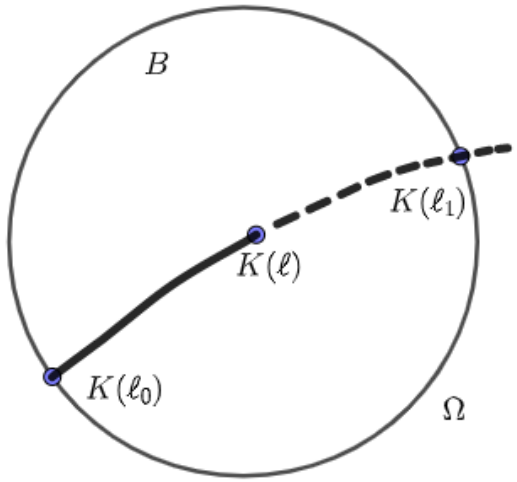

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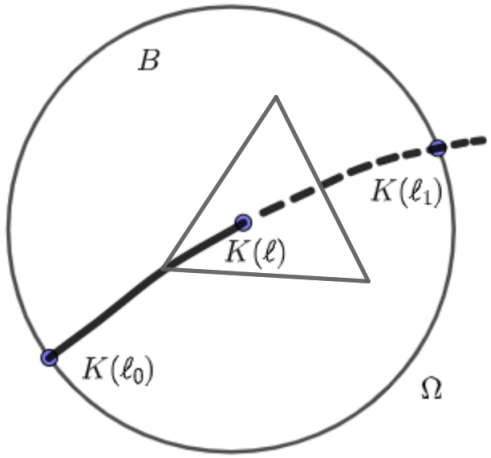

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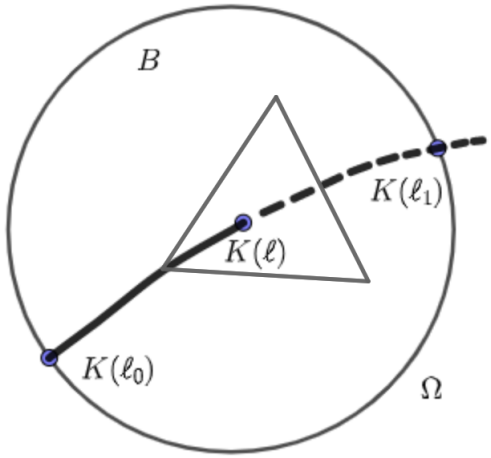

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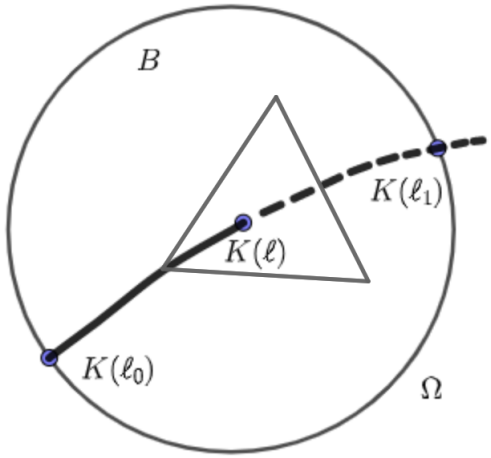

$$u = \frac{\kappa}{\mu} r^{\frac{1}{2}} \cos\frac{\theta}{2} + u^R, \qquad u^R \in H^2 \cap H^{1,\infty}.$$

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Substituting $u = \frac{\kappa}{\mu} r^{\frac{1}{2}} \cos\left(\frac{\theta}{2}\right) + u^R$ yields

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- $F_s(x) := \mathrm{I}(x) + \ell(s)\, V(x)$, diffeomorphism for small s.

$$G = \mu \int_{\Omega_0} \left[\frac{(\partial_1 u)^2 - (\partial_2 u)^2}{2} \left(\partial_1 V_1 - \partial_2 V_2\right) + \partial_1 u\, \partial_2 u \left(\partial_2 V_1 + \partial_1 V_2\right) \right] \mathrm{d}x$$

Substituting $u = \frac{\kappa}{\mu} r^{\frac{1}{2}} \cos\left(\frac{\theta}{2}\right) + u^R$ yields

$$G = \frac{\kappa^2}{2\,\mu}.$$

$$\frac{\mathrm{d}}{\mathrm{d}\ell} E\big(g, K(\ell)\big) = \underbrace{\gamma}_{= G_c} - \underbrace{\frac{\kappa\big(u(\ell), \ell\big)^2}{2\,\mu}}_{= G}.$$

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Griffith's criteria 2.0

for a.e. $t \in (0, 1)$

$$\begin{cases} \dot{\ell}(t) \geq 0 \\ \gamma - \dfrac{\kappa\Big(u\big(\ell(t)\big), \ell(t)\Big)^2}{2\,\mu} \geq 0 \\ \left\{\gamma - \dfrac{\kappa\Big(u\big(\ell(t)\big), \ell(t)\Big)^2}{2\,\mu}\right\} \dot{\ell}(t) = 0 \end{cases}$$

J-INTEGRAL AND REGULARITY OF κ

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J-integral and regularity of κ

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This can be used to prove that

$\kappa^2\big(u(\ell),\ell\big)$ is continuous w.r.t. the Hausdorff convergence of $K(\ell)$.

Thank you for your attention!

1. Historical remarks on LEFM
   - 1.1 Griffith's criterion
   - 1.2 Irwin's criterion
   - 1.3 J-integral
2. Variational formulation
   - 2.1 Francfort-Marigo model
   - 2.2 Interplay of energies
   - 2.3 Minimization over admissible cracks
   - 2.4 Strengths, weaknesses and extensions
   - 2.5 Our objective?
3. Two-dimensional anti-plane linear elasticity
   - 3.1 Assumptions and simplifications
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4. Preliminaries and overview of proof
   - 4.1 Deny-Lions spaces
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   - 4.3 Hausdorff distance and lower semi-continuity
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   - 5.1 Irwin's singularity
   - 5.2 Griffith's energy release rate
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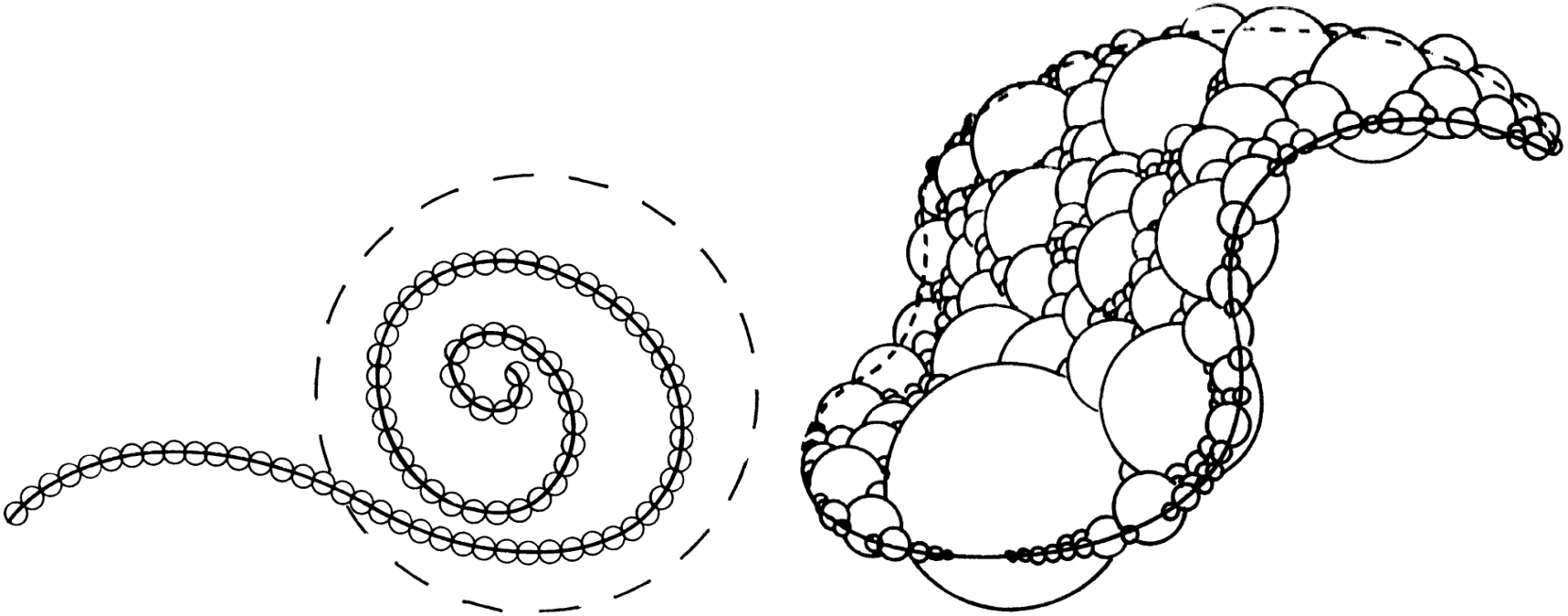
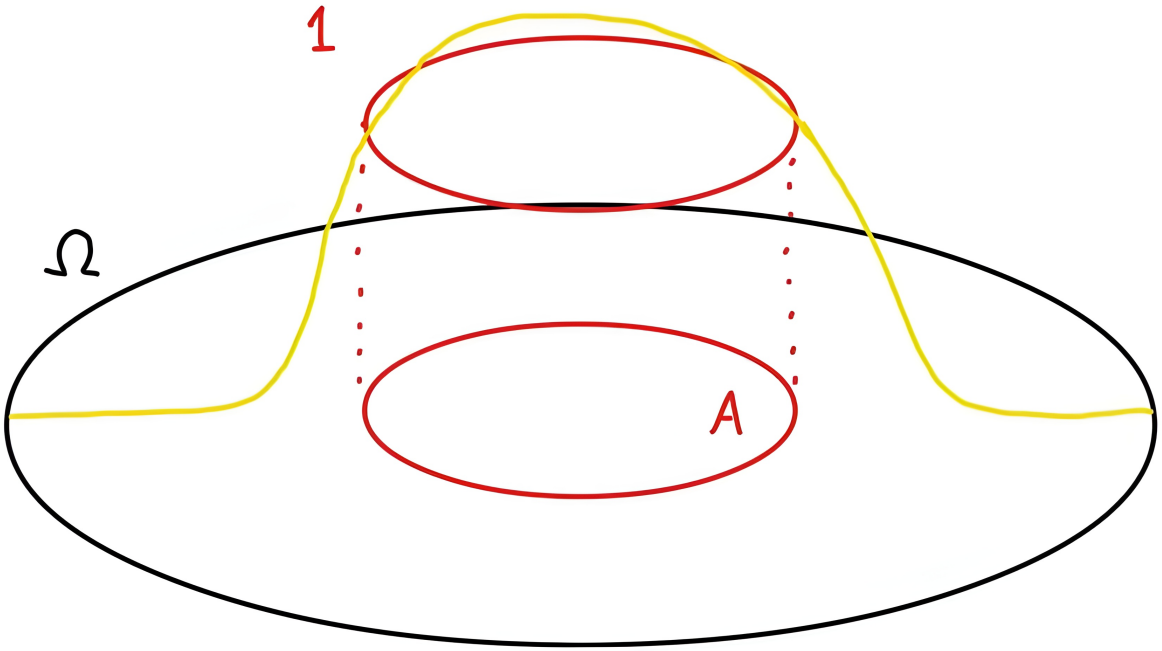
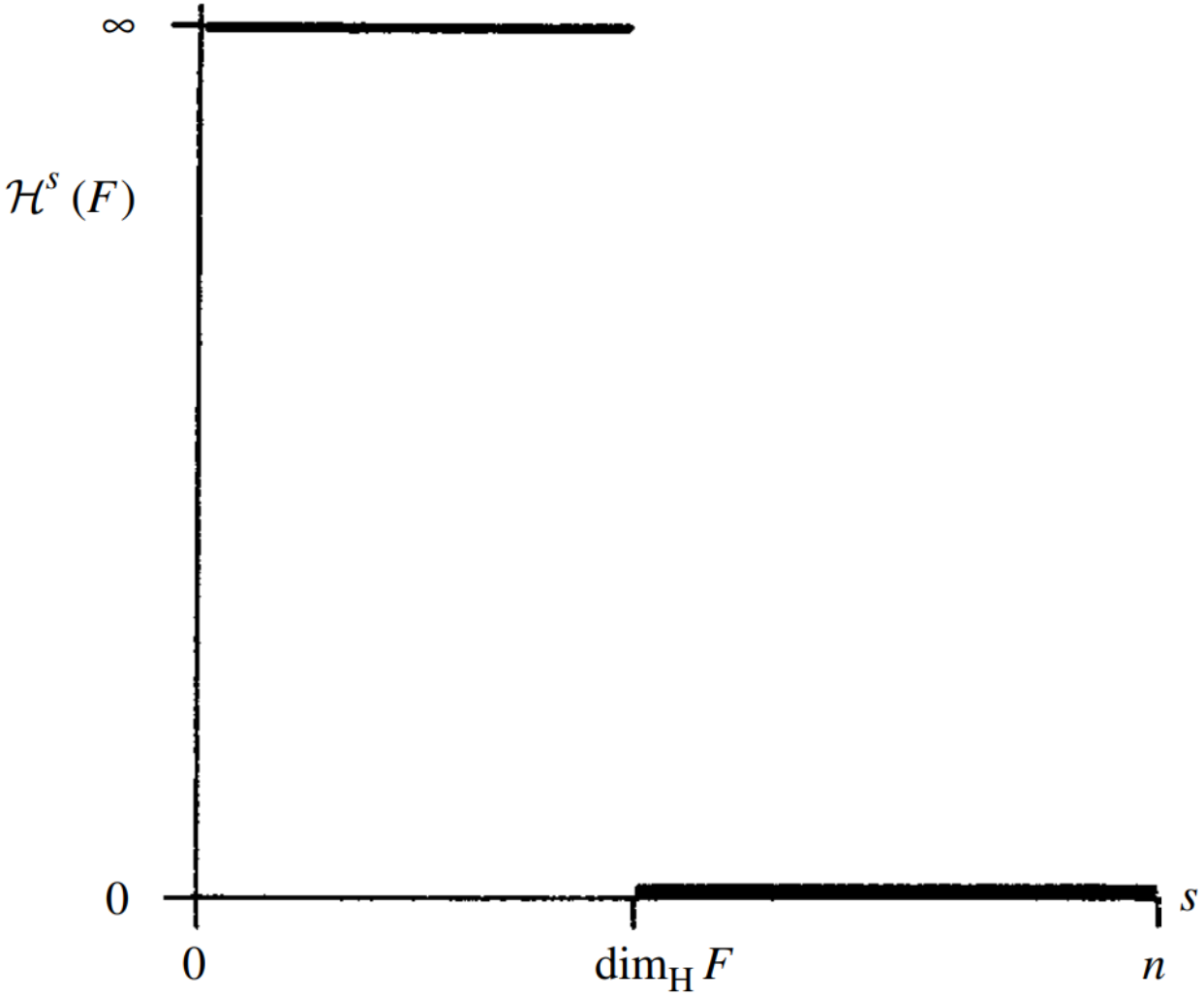
Hausdorff measure



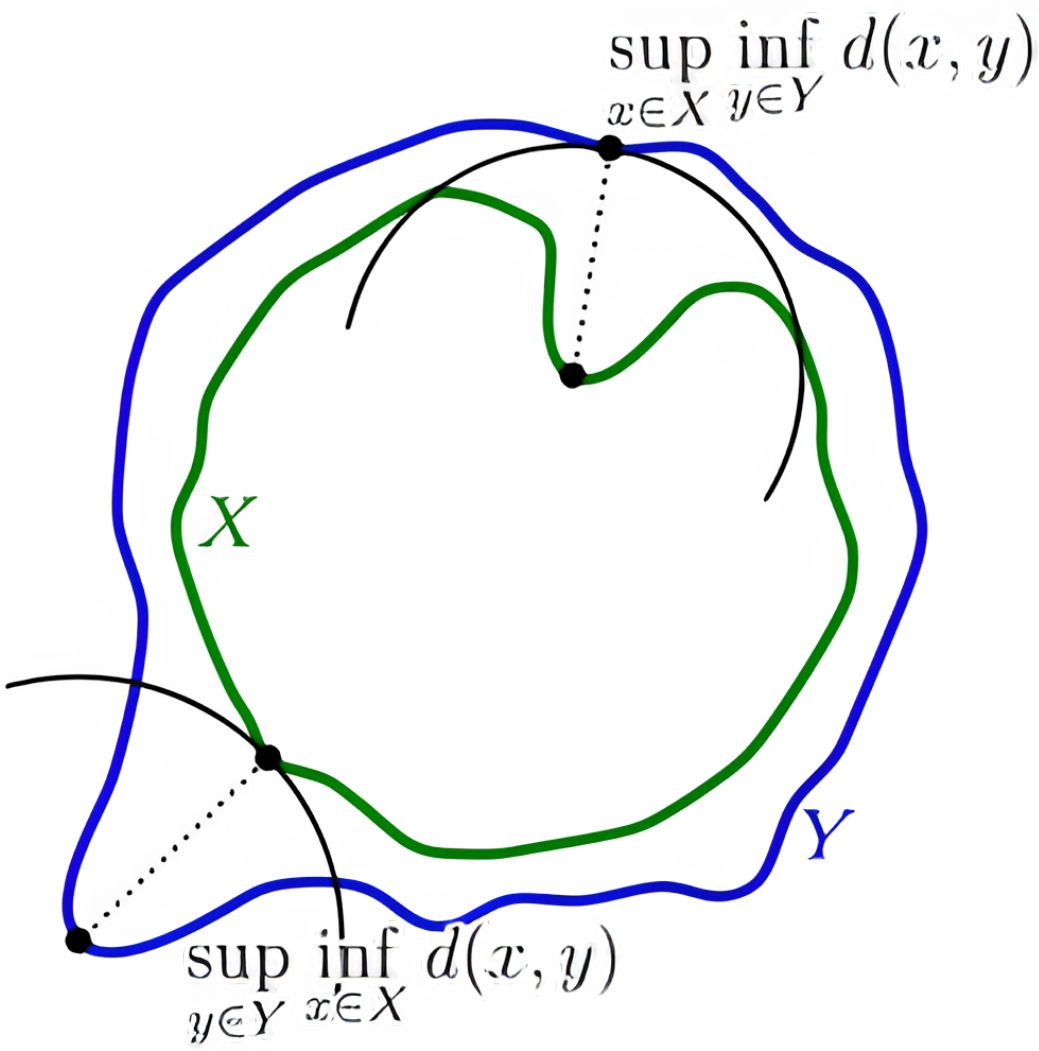
p-Capacity



Hausdorff dimension



Hausdorff distance



Ambrosio-Tortorelli functional

Our functional, in BV setting, would be:

$$E(u) = \frac{\mu}{2}\int_{\Omega\setminus J_u} |\nabla u|^2 \mathrm{d}x + \gamma\mathcal{H}^1(J_u)$$

An approximation was found by Ambrosio and Tortorelli (for a model of image segmentation, but it can be applied here also):

$$E_\varepsilon(u,v) = \frac{\mu}{2}\int_\Omega v^2|\nabla u|^2 \mathrm{d}x + \gamma\int_\Omega \left[\varepsilon|\nabla v|^2 + \frac{(v-1)^2}{4\varepsilon}\right]\mathrm{d}x$$

Then $E_\varepsilon(u,v)$ $\Gamma(L^2)$-converges to $E(u)$, and if we denote by $(u_\varepsilon, v_\varepsilon)$ and (u,v) the respective minimums, we have that $u_\varepsilon \to u$ in L^2, and $v_\varepsilon \to 1$ for $\mathcal{H}^1$-a.e. $x \in J_u$.